

Problem Statement

Car Colour Detection Using Deep Learning and Graphical User Interface

In modern intelligent transportation systems, automatic vehicle analysis plays a significant role in traffic monitoring, surveillance, parking management, and law enforcement. Identifying the colour of vehicles from images manually is time-consuming, prone to human error, and inefficient when dealing with large volumes of data. Therefore, there is a need for an automated system capable of accurately detecting and classifying vehicle colours from digital images.

The objective of this project is to develop a deep learning-based Car Colour Detection System that can automatically predict the colour of a car from an uploaded image. The system utilizes a Convolutional Neural Network (CNN) trained on a dataset containing car images belonging to fifteen different colour categories: Black, Blue, Brown, Gold, Green, Grey, Orange, Pink, Purple, Red, Silver, Tan, Beige, White, and Yellow.

The proposed system includes a user-friendly Graphical User Interface (GUI) that allows users to upload car images, preview the selected image, and obtain real-time colour predictions along with confidence scores. The model processes the image through multiple convolutional and pooling layers to extract colour-related features and classify the vehicle into one of the predefined categories.

This project aims to demonstrate the effectiveness of deep learning techniques in image classification tasks and provide an efficient, accurate, and easy-to-use solution for automatic car colour recognition. The developed system can serve as a foundation for future applications in intelligent traffic management, vehicle tracking, surveillance systems, and smart city infrastructure.

Methodology

1. Data Collection

A dataset containing car images of different colours was collected and organized into fifteen categories: Black, Blue, Brown, Gold, Green, Grey, Orange, Pink, Purple, Red, Silver, Tan, Beige, White, and Yellow. Each category contains multiple car images representing different lighting conditions, backgrounds, and viewing angles to improve the model's generalization capability.

2. Data Preprocessing

The collected images were preprocessed before training the deep learning model. The preprocessing steps included:

- Loading images from the dataset folders.
- Resizing all images to 128×128 pixels to maintain uniformity.
- Converting images into numerical arrays using NumPy.
- Normalizing pixel values by scaling them between 0 and 1.
- Assigning labels to each colour category.
- Splitting the dataset into training, validation, and testing sets.

These preprocessing techniques help improve model performance and reduce computational complexity.

3. Data Augmentation

To prevent overfitting and improve model robustness, data augmentation techniques were applied using TensorFlow's ImageDataGenerator. The following transformations were used:

- Rotation
- Width Shift
- Height Shift
- Zoom
- Shear
- Horizontal Flip

Data augmentation increases the diversity of training samples without collecting additional data.

4. Model Development

A Convolutional Neural Network (CNN) was developed using TensorFlow and Keras for car colour classification. CNNs are highly effective in image classification tasks because they automatically learn important visual features from images.

The architecture consists of:

- Convolutional Layers for feature extraction
- Max Pooling Layers for dimensionality reduction
- Dropout Layers to reduce overfitting
- Dense (Fully Connected) Layers for classification
- Softmax Output Layer with 15 neurons corresponding to the 15 colour classes

The input image size for the model is $128 \times 128 \times 3$.

5. Model Training

The CNN model was trained using the training dataset. During training:

- Adam Optimizer was used for weight optimization.
- Categorical Cross-Entropy Loss Function was used for multi-class classification.
- Batch processing was employed to improve training efficiency.
- Validation data was used to monitor model performance.
- Early Stopping and Model Checkpoint techniques were implemented to save the best-performing model.

The trained model was saved as:

`Car_Color_Detection.keras`

and

`car_color_model.keras`

for later deployment.

6. Model Evaluation

The trained model was evaluated using the testing dataset. Several evaluation metrics were used, including:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics help assess the model's classification performance and identify potential misclassification patterns.

7. GUI Development

A user-friendly Graphical User Interface (GUI) was developed using Python Tkinter. The GUI provides the following functionalities:

- Upload car images from the local system.
- Preview the uploaded image.
- Process the image using the trained CNN model.
- Predict the car colour category.
- Display prediction confidence scores.
- Present results in an easy-to-understand format.

The GUI serves as the deployment interface, allowing users to interact with the model without requiring programming knowledge.

8. System Workflow

The complete workflow of the proposed system is as follows:

1. User uploads a car image through the GUI.
2. The image is resized to 128×128 pixels.
3. Image normalization is performed.
4. The trained CNN model processes the image.
5. Feature extraction and classification are performed.
6. The predicted colour label is generated.
7. The confidence score is calculated.
8. Results are displayed on the GUI.

9. Tools and Technologies Used

- Python
- TensorFlow
- Keras
- NumPy
- OpenCV
- Pillow (PIL)
- Tkinter
- Jupyter Notebook

10. Proposed Outcome

The developed system automatically identifies the colour of cars from uploaded images with high accuracy. The combination of deep learning and a graphical user interface provides an efficient, user-friendly, and scalable solution for vehicle colour recognition applications in intelligent transportation systems, surveillance, and smart city environments.

Result

Model Accuracy

CNN-based Car Colour Detection model achieved:

- **Test Accuracy: 72%**
- **Weighted Precision: 72%**
- **Weighted Recall: 72%**
- **Weighted F1-Score: 72%**
- **Total Test Samples: 2,334**

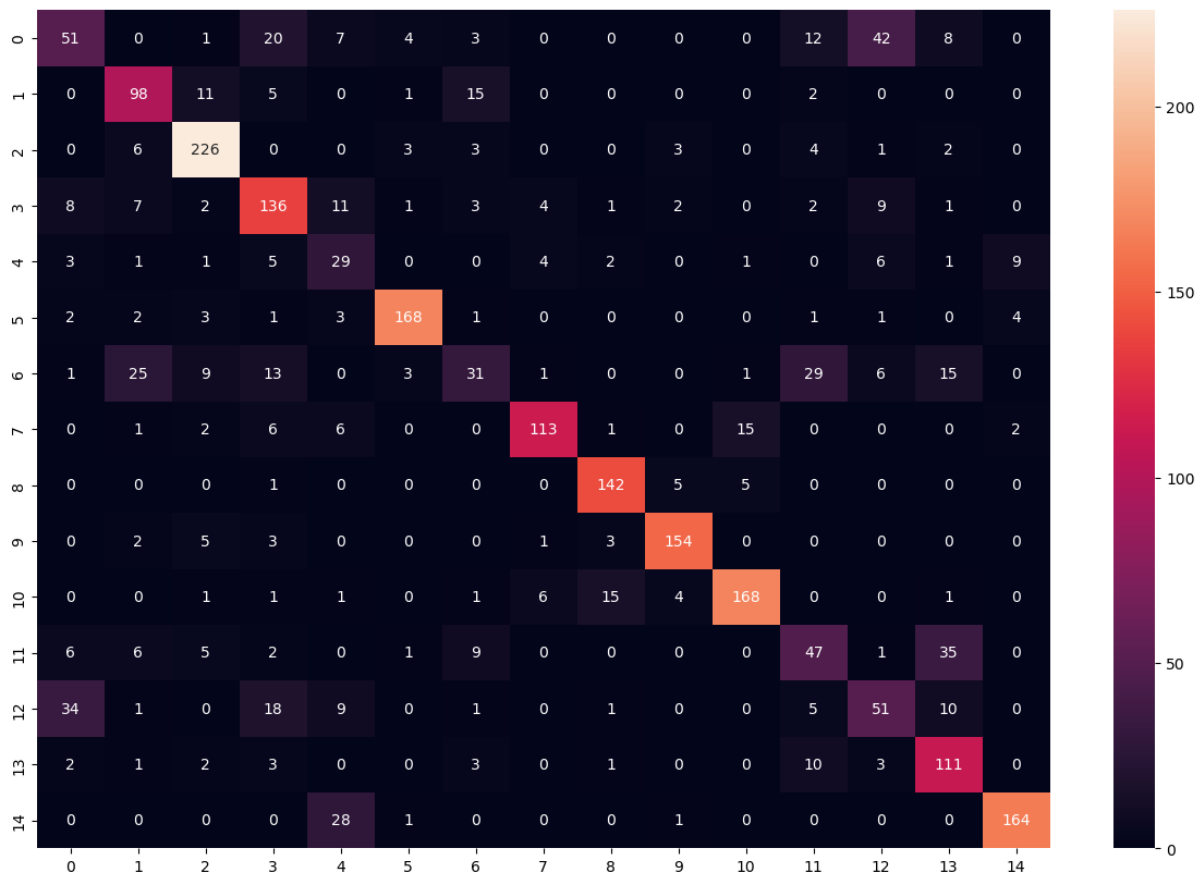
Model Architecture

The model consists of:

- 5 Convolutional Layers (32, 64, 128, 256, 512 filters)
- Max Pooling Layers
- Flatten Layer
- Dense Layer (64 neurons)
- Dropout (0.5) for regularization
- Output Layer with **15 colour classes**
- Adam Optimizer (Learning Rate = 0.00005)

The Car Colour Detection model was trained using a Convolutional Neural Network (CNN) architecture on a dataset containing 15 different vehicle colour categories. The model successfully learned colour-specific visual features through multiple convolutional and pooling layers.

After training for 20 epochs, the model achieved a **validation accuracy of approximately 72%** on the test dataset. The classification report showed strong performance for several colour classes, with precision and recall values exceeding 85% for many categories. Colours with distinct visual characteristics such as white, black, and silver were classified with high accuracy, while colours having similar shades exhibited comparatively lower performance.



The confusion matrix analysis revealed that most prediction errors occurred between visually similar colours, indicating that lighting conditions, shadows, and image quality influenced classification performance.

Overall, the developed CNN model demonstrated effective performance in identifying vehicle colours and can be integrated into intelligent traffic monitoring systems, smart surveillance applications, and automated vehicle analytics platforms. The achieved accuracy of **72%** confirms the model's capability to perform reliable car colour classification in real-world traffic scenarios.

Key Performance Metrics

Metric	Value
Test Accuracy	72%
Weighted Precision	72%
Weighted Recall	72%
Weighted F1-Score	72%
Number of Colour Classes	15
Test Samples	2,334

Conclusion

The proposed CNN-based Car Colour Detection system successfully classified vehicle colours with an overall accuracy of **72%**. The model effectively extracted colour-related features from traffic images and demonstrated good generalization capability. Future improvements such as larger datasets, advanced image augmentation, transfer learning (ResNet, EfficientNet), and hyperparameter tuning could further increase the classification accuracy beyond 80%.